MEDIUM

GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

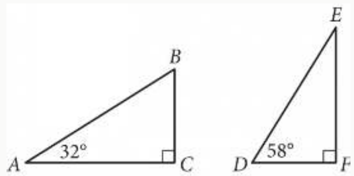
05

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GEOMETRY AND TRIGONOMETRY



Triangles ABC and DEF are shown above. Which of the following is equal to the ratio $\frac{BC}{AB}$?

A. $\frac{DE}{DF}$

B. $\frac{DF}{DE}$

C. $\frac{DF}{EF}$

D. $\frac{EF}{DE}$

Q2

GEOMETRY AND TRIGONOMETRY

A sphere has a radius of $\frac{17}{5}$ feet. What is the volume, in cubic feet, of the sphere?

A. $\frac{5\pi}{17}$

B. $\frac{68\pi}{15}$

C. $\frac{32\pi}{5}$

D. $\frac{19{,}652\pi}{375}$

Q3

GRID-IN GEOMETRY AND TRIGONOMETRY

An angle has a measure of $\frac{16\pi}{15}$ radians. What is the measure of the angle, in degrees?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4 GEOMETRY AND TRIGONOMETRY

In $\triangle ABC$, $\angle B$ is a right angle and the length of $\bar{BC}$ is 136 millimeters. If $\cos A = \frac{3}{5}$, what is the length, in millimeters, of $\bar{AB}$?

A. 34

B. 102

C. 136

D. 170

Q5 GEOMETRY AND TRIGONOMETRY

Two nearby trees are perpendicular to the ground, which is flat. One of these trees is 10 feet tall and has a shadow that is 5 feet long. At the same time, the shadow of the other tree is 2 feet long. How tall, in feet, is the other tree?

A. 3

B. 4

C. 8

D. 27

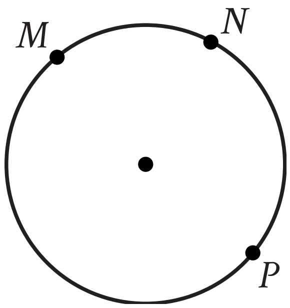
The length of the edge of the base of a right square prism is 6 units. The volume of the prism is 2,880 cubic units. What is the height, in units, of the prism?

A. $4\sqrt{30}$

B. 36

C. $24\sqrt{5}$

D. 80



- Clockwise from top left, the following points are on the circle:
    - upper M
    - upper N
    - upper P

Points M, N, and P lie on the circle shown. On this circle, minor arc MN has a length of 39 centimeters and major arc MPN has a length of 195 centimeters. What is the circumference, in centimeters, of the circle shown?

A. 39

B. 156

C. 195

D. 234

Q8

GEOMETRY AND TRIGONOMETRY

One leg of a right triangle has a length of 43.2 millimeters. The hypotenuse of the triangle has a length of 196.8 millimeters. What is the length of the other leg of the triangle, in millimeters?

A. 43.2

B. 120

C. 192

D. 201.5

Q9

GEOMETRY AND TRIGONOMETRY

Each side of equilateral triangle S is multiplied by a scale factor of k to create equilateral triangle T. The length of each side of triangle T is greater than the length of each side of triangle S. Which of the following could be the value of k?

A. $\frac{29}{28}$

B. 1

C. $\frac{28}{29}$

D. 0

Q10

GEOMETRY AND TRIGONOMETRY

Circle K has a radius of $4 \text{ millimeters } (\text{mm})$. Circle L has an area of $100\pi \text{ mm}^2$. What is the total area, in mm^2, of circles K and L?

A. 14π

B. 28π

C. 56π

D. 116π